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Experimental Study on the Impact of Hydrogen Peroxide on the Alteration of Shale Properties During the Oxidative Dissolution Process

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Abstract

Gas extraction from shale is challenging due to the extremely low permeability of shale formations. In response, an oxidative dissolution treatment method was introduced to eliminate organic matter (kerogen) present in shale rock, enhance pore diameter, and increase the permeability of the formation. This research study investigates the potential of hydrogen peroxide (H_2O_2) as an effective oxidizing agent for enhancing the properties of shale cores.

In detail, the impact of the oxidative dissolution process on the characteristics of shale cores was assessed utilizing two chemical systems: H_2O_2 alone and a slick water recipe (containing several additives and H_2O_2) with concentrations of 30% H_2O_2. Four shale cores were separately immersed in both solutions for durations of 2 and 4 hours at a temperature of 120°F. Various characterization techniques were employed to detect the alterations in core properties subsequent to the introduction of the chemical systems.

The oxidative treatment of Eagle Ford shale cores utilizing H_2O_2 and the slick water system resulted in considerable alterations in surface morphology, porosity, and chemical composition. The SEM analysis indicated a significant enhancement in micro-porosity and surface etching, particularly following 4 hours of treatment at 120 °F, which suggests the progressive oxidative decomposition of kerogen within the Eagle Ford shale matrix. Complementary EDX data corroborated a substantial reduction in carbon content, decreasing from approximately 14.2 wt.% in untreated cores to 7.8 wt.% and 6.1 wt.% after treatment with H_2O_2 alone and the slick water formulation, respectively. Photographic images further substantiated these findings, demonstrating notable darkening and textural roughness post-treatment, as well as visual indicators of surface oxidation and the removal of organic matter. Gravimetric analysis revealed consistent weight loss across cores, ranging from 0.006 to 0.030 grams per core, thus verifying the partial dissolution of organic content without compromising core structural integrity. TOC analysis via pyrolysis further affirmed these trends, with TOC values declining from 5.7 wt.% in untreated cores to 3.2 wt.% and 2.6 wt.% following treatment with H_2O_2 and slick water, respectively. Measurements of Young's modulus by AutoScan indicated a reduction in rock stiffness ranging from 30% to 60%, signifying an increase in fractur ability following treatment.

Collectively, these findings underscore the viability of H_2O_2-based oxidative strategies, especially when augmented with additive-rich slick water systems, as effective methodologies for enhancing pore network development and gas production in tight shale formations. Future investigations should explore the interplay between organic matter removal and secondary phenomena such as mineral scaling, precipitation, and alterations in thermal stability, all of which may impact long-term permeability and hydrocarbon recovery efficiency.

Introduction

The demand for the exploration and development of shale oil and gas has been rapidly increasing, attracting significant attention. Shale gas, recognized as a significant unconventional natural gas resource, has garnered increased interest recently. The global total of shale gas resources is estimated to be approximately 4.56 × 10^14 m^3 (Perry & Lee, 2007). To exploit these vast reserves, a variety of methods and technologies, including horizontal drilling and hydraulic fracturing, have been employed to enhance gas recovery (Fatahi et al., 2016; Yang et al., 2016; Ma et al., 2017). However, the primary challenge in shale gas exploration lies in the extremely low porosity and permeability of shale formations. Furthermore, the associated environmental issues, such as water pollution, must also be taken into consideration (Ziemkiewicz & Thomas He, 2015). Consequently, it is essential to develop innovative, environmentally friendly, and more effective technologies while simultaneously researching their environmental impacts.

Research has shown that carbon dioxide (CO_2) is a suitable candidate for displacing methane due to its superior adsorption capacities for shale gas extraction (Weniger et al., 2010; Lin et al., 2016). Consequently, CO_2, including CO_2-based fluids and supercritical CO_2, has gradually been used as a fracturing fluid for shale gas extraction (Merey & Sinayuc, 2016; Yin et al., 2016; Zhang et al., 2017). However, several technical challenges associated with CO_2 fracturing must be addressed, including pipeline issues, the phase change of CO_2, security concerns, and economic feasibility. Therefore, both academic and industrial professionals are actively seeking alternative solutions for shale gas extraction.

On the other hand, extensive analyses of micro-pore structures have been conducted to investigate the mechanisms of gas adsorption. Preliminary research indicates that methane adsorption is intimately connected to organic matter and the specific surface area. According to Li et al., the adsorption capacity of shale gas exhibits a positive correlation with organic matter, which can provide a substantial specific surface area (Cao et al., 2016). Firoozabadi et al. discovered that methane adsorption primarily depends on surface area, with adsorption predominantly occurring near the shale surface as pore size increases (Jin & Firoozabadi, 2013). Additionally, Zhang et al. established a linear correlation between methane adsorption capacity and specific surface area in clay-mineral dominated rocks. Methane molecules preferentially occupy the surface sites of organic matter (Ji et al., 2012).

Furthermore, due to the blockage of pores by bituminous kerogen, organic matter significantly influences the specific surface area, thus restricting the development of nanopores and gas adsorption (Saidian et al., 2016). Therefore, removing organic matter through oxidation can increase reservoir permeability for gas recovery, making the process technically and economically feasible. For example, eliminating organic matters using high-temperature pyrolysis and combustion resulted in an increase in the pore volume and specific surface area (Sun et al., 2015). In addition to high-temperature oxidation, different chemical oxidizer solutions were also used for shale treatment at ambient temperature. Zhou et al. used dichloromethane to remove extractable organic matter. Then, they used H_2O_2 to remove solid organic matter, thus increasing the pore volume and porosity of the shale (Li et al., 2016). Supercritical CO_2 solution and buffered sodium hypochlorite (NaOCl) solution were also used to remove organic matters in the shale (Kuila et al., 2014). The degree of roughness of the shale's geometrical surface decreased, and the morphology of the shale pore structure transformed gradually from complex to regular (Yin et al., 2016).

In contrast to the aforementioned chemical oxidizer solutions, hydrogen peroxide (H_2O_2) is a cost-effective oxidizer, and its decomposition products (H_2O and O_2) pose no potential hazardous effects on the environment. H_2O_2 can be employed as an oxidative additive to remove kerogen from shale, thus increasing its permeability and potentially enhancing hydrocarbon recovery. H_2O_2 reacts with kerogen, facilitating its breakdown and creating new pore spaces and pathways within the shale. This process may be particularly effective when combined with other techniques, such as hydraulic fracturing. H_2O_2 has already been utilized to enhance coal quality (Levent et al., 2007; Jin et al., 2011; Liu et al., 2013; Doskocil et al., 2014). However, studies investigating the use of H_2O_2 as an oxidizer to react with organic matter and enlarge the pore diameter of shale have been reported by a limited number of researchers.

This study aims to evaluate the impact of H_2O_2 on the alterations in the properties of shale cores. A comprehensive analysis will be undertaken to elucidate the impact of the reaction time of H_2O_2, whether utilized independently or as part of a slick water recipe, on the morphological, elemental, mechanical, and total organic content (TOC) of the tested shale cores via accurately quantify the mineral content utilizing SEM, EDX, and AutoScan techniques.

Materials

Four Eagle Ford shale cores, characterized by their black color, were meticulously shaped into cubic forms with dimensions of 2.52 × 1.3 cm^2 and subsequently utilized in this study. These cores were categorized into two distinct groups: the first group comprised two cores, which were treated individually with 30% H_2O_2 at 120 °F for durations of 2 and 4 hours. Conversely, the second group was subjected to the slick water system, which incorporates various additives, including H_2O_2, at a temperature of 120 °F for respective periods of 2 and 4 hours. The H_2O_2 utilized in this study contained a concentration of 30 wt% to treat the initial group of cores. The slick water system was prepared by combining several additives with H_2O_2 at 77°F for several minutes. Subsequently, the second group of cores was treated at 120 °F for the above-mentioned intervals of 2 and 4 hours.

Methods

It is widely acknowledged that reservoir conditions fluctuate in relation to burial depth (Liu et al., 2016). For instance, reservoir temperature escalates with the incremental depth of rocks (e.g., geothermal gradients) and is significantly higher than the ambient temperature. This article presents a preliminary investigation into the reaction mechanism of organic matter with H_2O_2, through immersion tests conducted at a temperature of 120°F.

The four shale cores were initially characterized using several techniques, including Scanning Electron Microscopy (SEM), X-Ray Diffraction (XRD), Energy Dispersive X-Ray Analysis (EDX), and AutoScan. Subsequently, each individual core was immersed in the treatment solution, comprising either 30% H_2O_2 or the slick water system, within a laboratory water-bath apparatus at predetermined concentrations, as detailed in Table 1. The treatment duration for each core was predetermined to be either 2 hours or 4 hours, conducted at a temperature of 120°F. Following the treatment, the shale cores were subjected to a careful washing process utilizing deionized water to eradicate impurities. Subsequently, they were dried in an oven set to a specified temperature of 80 °F for 24 hours.

Table 1—The oxidative treatment conditions involve applying either 30% H_2O_2 or a slick water system using a laboratory water-bath apparatus:

Shale Core	Temperature (°F)	Oxidative Fluid Type	Time (Hours)
(a)	120	30 % H_2O_2	2
(b)		30 % H_2O_2	4
(c)		Slick water	2
(d)		Slick water	4

Thereafter, the properties of the four treated cores were re-evaluated utilizing SEM, X-Ray, EDX, and AutoScan. Furthermore, the weight loss of the shale cores was measured using a standard laboratory balance before and after the immersion step. The pyrolysis method was employed in the absence of oxygen in order to analyze the core's TOC. The AutoScan tests were conducted to assess the alteration in mechanical properties before and after applying the chemical treatment to the shale cores. These evaluations were performed to determine the impacts of the oxidative dissolution treatment under designated conditions on the characteristics of the shale cores that were subjected to testing. Fig. 1 illustrates the experimental methodology employed to assess the impact of oxidative dissolution on shale cores.

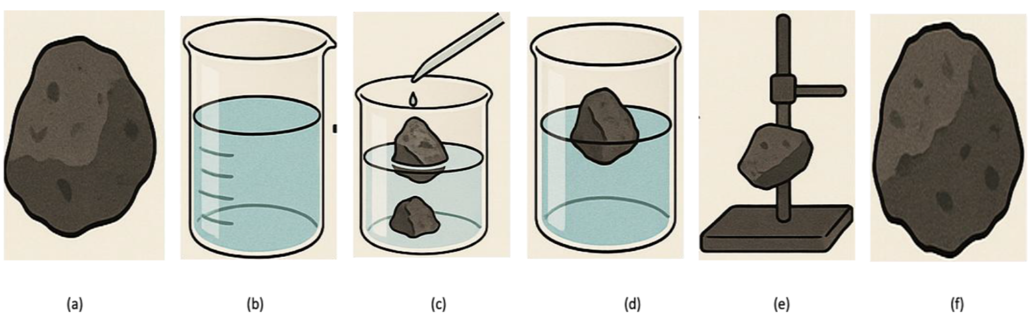

Figure 1—The experimental workflow design of oxidative dissolution involving H_2O_2 and the slick water system on the assessed shale cores. The sequence of steps is as follows: (a) Select and pre-characterize the shale core; (b) Prepare the oxidative treatment systems; (c) Immerse the shale cores in the oxidative solution; (d) Wash the treated cores with deionized water; (e) Dry the washed cores; (f) Conduct post-treatment characterization.

Results And Discussion

Mechanism of transformation in shale cores under oxidative dissolution reactions

The mechanism through which the properties of shale cores are transformed following the application of oxidative dissolution utilizing H_2O_2 alone and in conjunction with a slick water system primarily relies on the deliberate removal of organic matter within the rock matrix. Shale formations frequently encompass substantial quantities of kerogen and bitumen encapsulated within nanopores and microfractures. The oxidative agents interact with this organic matter, decomposing complex hydrocarbons into soluble or volatile byproducts such as CO_2, H_2O, and low-molecular-weight compounds.

This chemical oxidation process serves multiple objectives, including: (a) enhanced porosity and microfracture generation as organic matter undergoes dissolution, previously occluded pore spaces are liberated, and new microfractures are established. This phenomenon results in a substantial enhancement in both matrix porosity and permeability. (b) The enhanced connectivity of pore networks, wherein the gradual dissolution promotes the linkage of isolated pore throats, facilitates the establishment of fluid flow pathways within the shale matrix. (c) The reduction in capillary pressure and flow resistance, achieved through the removal of viscous or immobile organic phases, diminishes fluid retention. This process consequently facilitates the migration of gas molecules toward production wells with greater ease. (d) Fracture bath expansion refers to the expanded microfracture regions that are formed as a result of oxidative attack. These regions function as localized areas of heightened gas accumulation and flow, serving as localized zones of increased gas accumulation and flow.

The oxidative dissolution process not only modifies the geochemical composition of the shale but also restructures its internal architecture, culminating in enhanced hydrocarbon recovery efficiency and elevated production rates. This technique presents a promising low-energy enhancement strategy for stimulating unconventional reservoirs.

The primary potential reaction mechanisms of H_2O_2 and shale cores can be classified into two categories (Mikutta et al., 2005): (i) H_2O_2 directly oxidizes organic matter in a peroxidic-type reaction. (ii) Specific metal ions present in minerals, such as Fe^{2+} or Ti^{3+}, exert a catalytic effect on the decomposition of H_2O_2, resulting in the formation of free radicals, including hydroxyl radicals (·OH) and hydroperoxyl radicals (HOO·), as elucidated in equations (1) and (2):

$$M^{2+} + H_2O_2 \rightarrow M^{3+} + OH^{-} + \cdot OH \tag{1}$$

$$M^{3+} + H_2O_2 \rightarrow M^{2+} + H^{+} + HOO. \tag{2}$$

Where M is a metal ion.

The hydroxyl radicals, characterized by their significant oxidation capacity, readily engage in electron transfer and abstract hydrogen atoms from other molecules, leading to the formation of water molecules (Gligorovski et al., 2015). As a result, hydroxyl radicals facilitate the decomposition of organic matter by reacting with carbon-carbon (C-C) bonds, carbon-hydrogen (C-H) bonds, carbon-carbon double (C=C) bonds, among others (Gligorovski et al., 2015). Through the removal of organic matter, the pore diameter will be enlarged, and an increased number of pores will be generated. Consequently, H_2O_2 may enhance the porosity.

Photographic and microstructural alterations in shale cores during oxidative dissolution reactions

The photographic comparison of untreated and treated Eagle Ford shale cores underscores the considerable surface alterations induced by oxidative treatment utilizing H_2O_2 and the slick water system, as demonstrated in Fig. 2. Prior to treatment, the shale cores exhibit a distinct bluish-gray coloration, characterized by well-preserved lamination and a relatively smooth, compact surface morphology. These attributes are emblematic of organic-rich shale comprising a tightly bound kerogen-mineral matrix. Subsequent to treatment, however, the images reveal a marked change in both coloration and texture. The cores assume a darker, more matte appearance, indicative of surface oxidation and the extraction of organic material. This darkening effect is particularly pronounced in cores treated with the slick water system, which incorporates multiple additives alongside H_2O_2, suggesting an enhanced oxidative efficiency and deeper penetration into the matrix. The observed loss of surface seen and augmented roughness in the post-treatment cores reflect microstructural alterations consistent with the decomposition of kerogen and the emergence of secondary porosity. These findings substantiate the effectiveness of oxidative systems in selectively targeting and modifying the organic framework within shale, potentially augmenting fluid transport pathways and gas recovery in unconventional reservoirs.

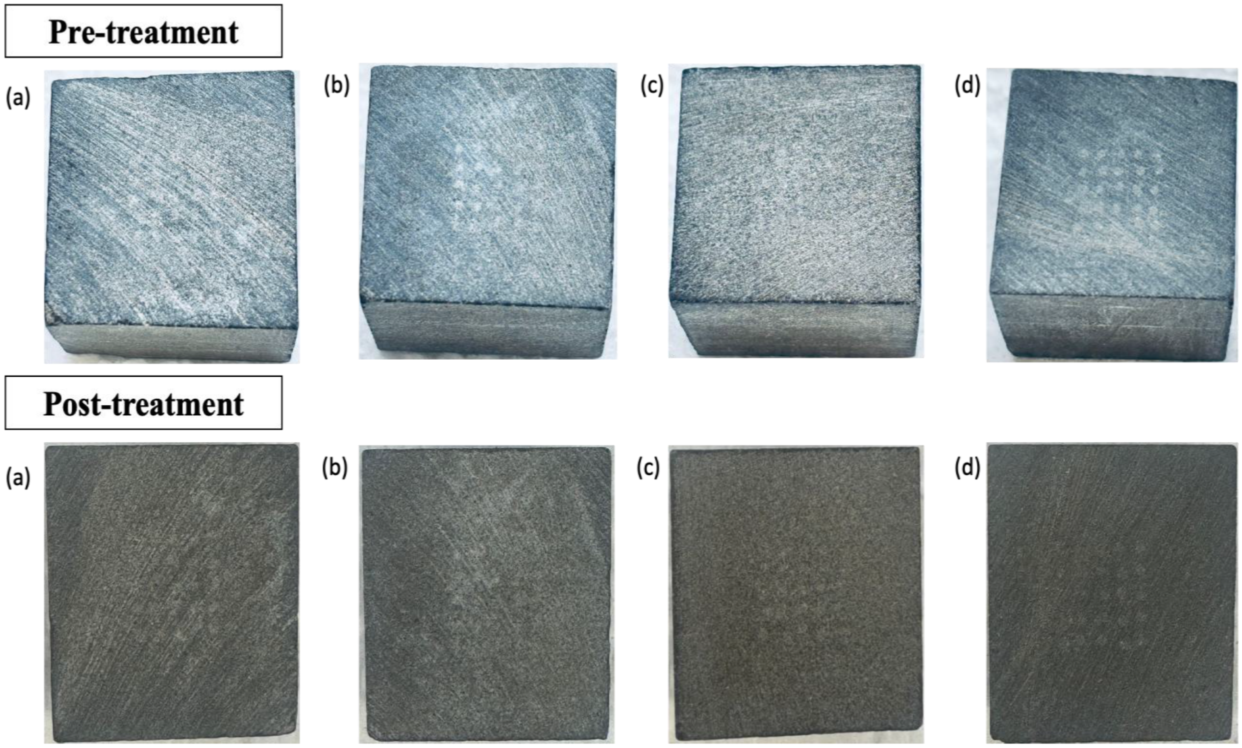

Figure 2—Above Photographic images of untreated Eagle Ford shale (a-d). Below images of core treated with (a) H_2O_2 for 2 hours, (b) H_2O_2 for 4 hours, (c) slick water for 2 hours, and (d) slick water for 4 hours.

On the other hand, the SEM imaging technique was employed to investigate the surface morphology and identify microstructural alterations in the shale cores at high magnification before and after the introduction of oxidative solutions at 120°F for durations of 2 and 4 hours. Fig. 3 provides a comprehensive overview of SEM images obtained from four untreated Eagle Ford shale cores to establish a baseline characterization of their microstructural features before the application of oxidative treatment. Each core exhibited a dense, compact matrix with varying distributions of organic-rich phases, predominantly kerogen, embedded within the fine-grained mineral framework. Core (a) displayed a relatively smooth surface with minimal pore development and extensive coverage of amorphous organic matter, indicating a low native porosity and limited natural permeability. In core (b), isolated organic inclusions and partially filled pore spaces were observed, indicating moderate kerogen saturation alongside early-stage natural microfracture features. Core (c) exhibited increased surface roughness and a slightly more advanced pore structure, encompassing nanoscale voids and fine cracks likely resulting from diagenesis or mechanical stress. Core (d) demonstrated the highest degree of natural heterogeneity, characterized by irregular organic clusters and minor secondary porosity, while still lacking significant interconnected pathways. Across all cores, kerogen appeared to dominate a substantial portion of the internal surface area, serving as a pore-filling material and obstructing potential flow channels. These observations provide essential structural context for understanding the initial condition of the shale matrix prior to oxidative decomposition, which will serve as a crucial reference point for evaluating morphological and porosity changes following H_2O_2-based kerogen removal treatments.

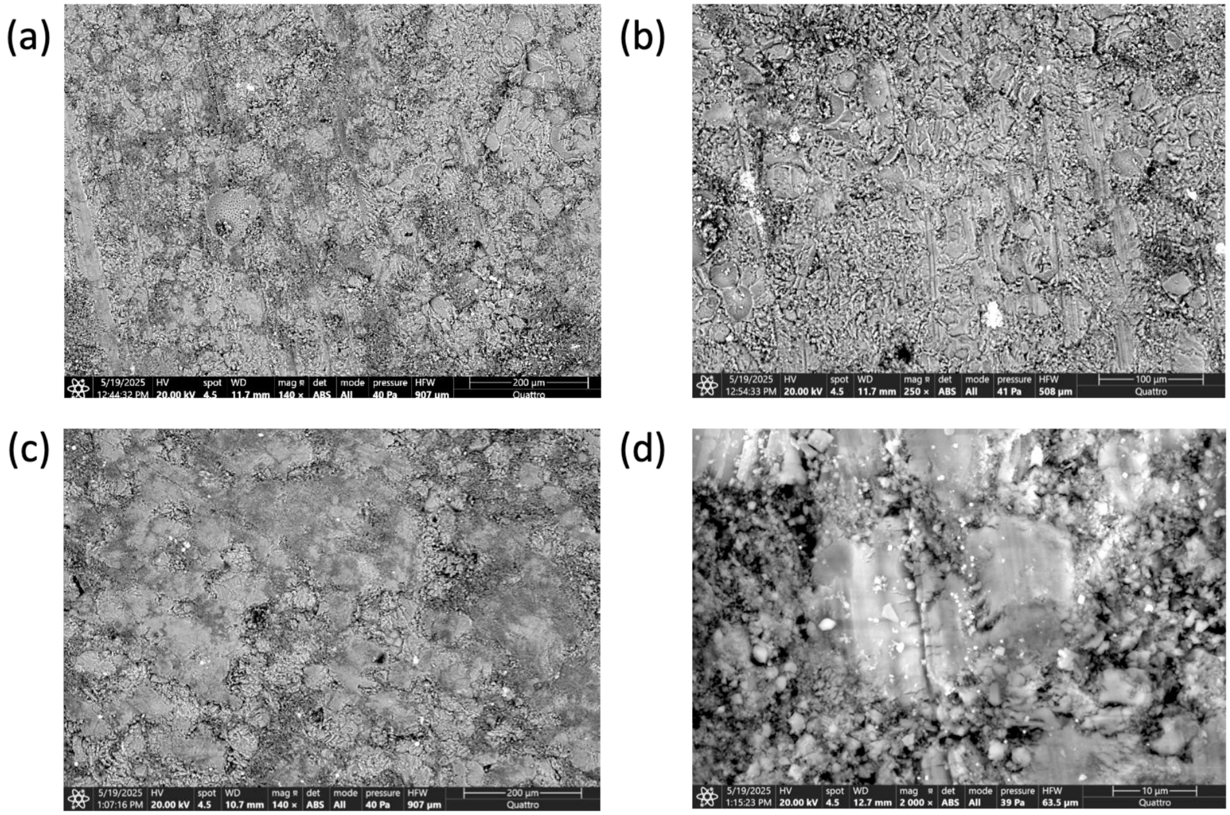

Figure 3—SEM images were obtained from four untreated Eagle Ford shale cores (a-c).

In contrast, SEM images demonstrated in Fig. 4 of Eagle Ford shale cores treated with oxidative fluids at 120 °F provide clear microstructural evidence of kerogen decomposition and porosity evolution under varying treatment conditions. Core (a) exposed to pure H_2O_2 for 2 hours exhibited the onset of surface etching and partial degradation of organic matter. The SEM micro-image revealed isolated nanoscale pits and a few early-stage micropores forming along kerogen-rich areas, while the majority of the mineral matrix remained intact. The reaction appeared surface-limited, exhibiting minimal development of connected porosity or microfractures. In contrast, following 4 hours of H_2O_2 exposure, the microstructure of the core (b) displayed deeper etching and substantial kerogen loss, particularly along bedding planes and organic laminae. Porosity increased through the emergence of interconnected pores and the appearance of secondary microfractures. However, evidence of grain boundary weakening was observed, as indicated by disrupted mineral contacts and initial signs of particle detachment, suggesting that extended exposure may jeopardize mechanical integrity.

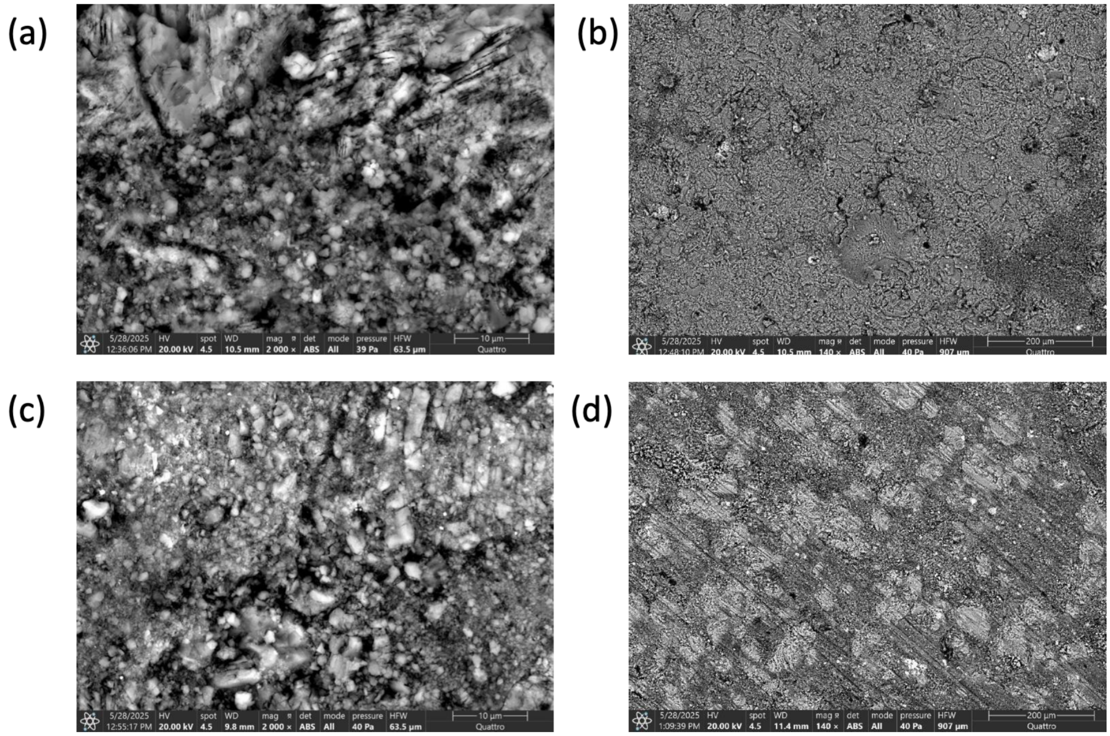

Figure 4—SEM Images of Eagle Ford shale cores subjected to oxidative fluids at a temperature of 120 °F include: core treated with (a) H_2O_2 for 2 hours, (b) H_2O_2 for 4 hours, (c) slick water for 2 hours, and (d) slick water for 4 hours.

In the case of the slick water system, involving a formulation that combines H_2O_2 with several additives designed to enhance oxidative efficiency and wettability. Core (c) was exposed to the system for 2 hours; the SEM images demonstrated markedly more advanced decomposition of organic phases in comparison to the pure H_2O_2 treatment. Organic matter was extensively removed from the surface, unveiling clean, sharp-edged pores and well-defined microcracks penetrating deeper into the matrix. The presence of additives appeared to catalyze more uniform pore development and early interconnectivity, likely through improved transport of reactive species into the shale's nanostructure. Following 4 hours of treatment with the slick water system, the microstructure of the core (d) underwent a dramatic transformation. The organic matter was almost entirely eliminated, resulting in a highly porous, fractured network with expanded pore diameters and coalesced voids. The SEM image revealed extensive surface roughening, grain lifting, and fracture propagation, particularly in areas that were formerly rich in kerogen. Although porosity and pore connectivity were greatly enhanced, certain regions exhibited evidence of over-etching, with partially collapsed pore walls and disaggregated mineral grains, suggesting that prolonged exposure to the enhanced oxidative system may compromise mechanical stability. These comparative observations confirm that both H_2O_2 and slick water treatments are effective in removing kerogen and increasing porosity in Eagle Ford shale cores; however, the slick water system achieves these results more expeditiously and uniformly due to synergistic additive effects. Nonetheless, the trade-off between aggressive organic removal and matrix preservation must be accurately balanced, as excessive treatment duration, even with optimized fluids, can result in undesirable structural degradation.

Mineral alterations in shale cores during oxidative dissolution reactions

Energy-Dispersive X-ray Spectroscopy (EDX) analysis spectra of untreated Eagle Ford shale cores (a-d) provide a comprehensive elemental profile that reflects the intricacies of the mineralogy and organic richness inherent in this formation prior to any chemical treatment. The EDX results are presented in Fig. 5, which affirm that the untreated Eagle Ford shale showcases a complex interaction of siliciclastic, carbonate, and organic constituents, with kerogen intricately associated with the mineral matrix. In details, The spectra consistently identified major elements, including silicon (Si), aluminum (Al), oxygen (O), calcium (Ca), iron (Fe), and carbon (C), along with trace amounts of magnesium (Mg), sulfur (S), potassium (K), and titanium (Ti). The prominent presence of Si and Al in high relative abundance is indicative of a clay-rich and quartz-bearing matrix, predominantly composed of illite, smectite, and kaolinite, in addition to fine-grained quartz silt. The pronounced O peak, in conjunction with Si and Al, reinforces the prevalence of aluminosilicate minerals. Ca is primarily associated with calcite ($CaCO_3$) or other carbonate phases such as dolomite, which are characteristic diagenetic components of the Eagle Ford formation. Iron, frequently found in pyrite (FeS_2) or iron oxides, suggests the existence of redox-sensitive mineral phases that may facilitate catalytic roles in oxidative reactions during subsequent treatments. C manifests in significant quantities, correlating with the shale's high TOC content and the presence of kerogen interspersed within the mineral matrix. The carbon signal, in conjunction with S and Fe, may also indicate the presence of organo-sulfur complexes or framboidal pyrite inclusions distributed within the organic matter. The Mg and K peaks further substantiate the existence of clay minerals, particularly illite-smectite mixed layers, while trace Ti probably originates from detrital rutile or ilmenite. This foundational compositional data is essential for evaluating prospective chemical modifications resulting from oxidative treatment, particularly in monitoring the depletion of organic carbon and the evolution of reactive mineral phases under H_2O_2-based fluid systems.

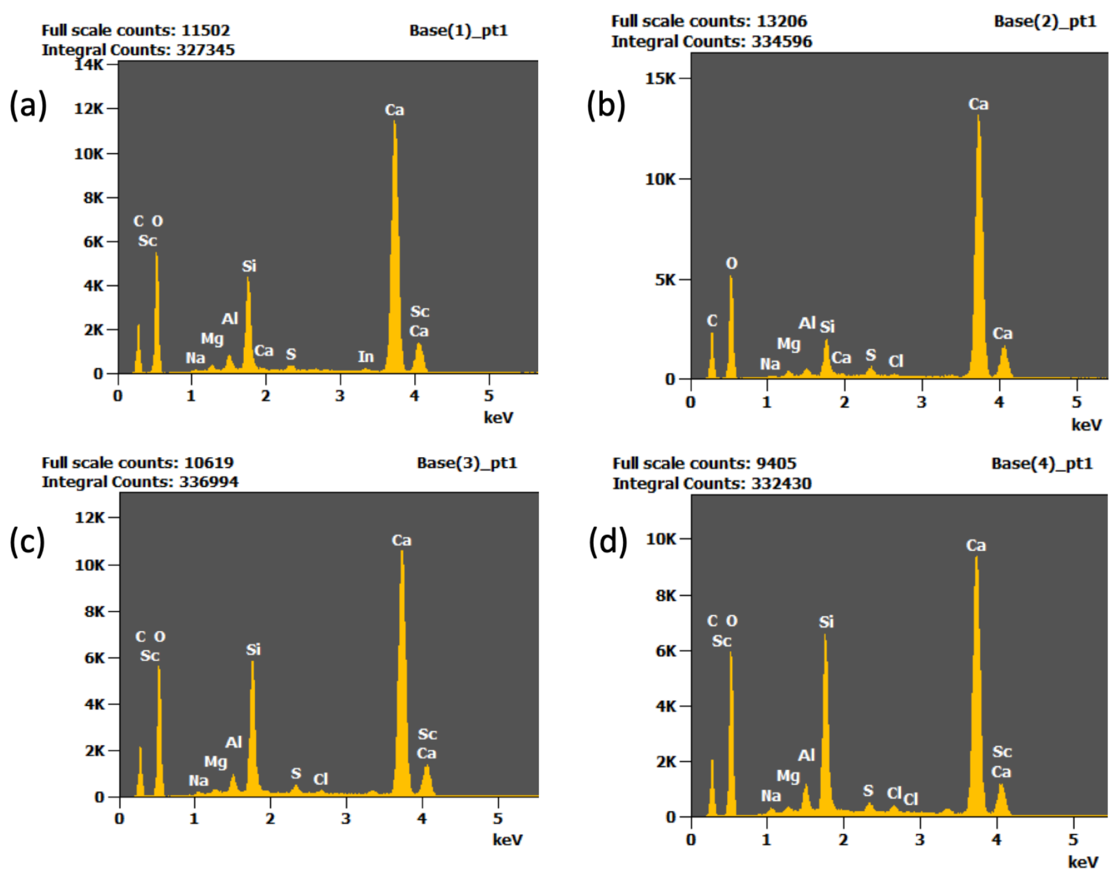

Figure 5—EDX analysis spectra of untreated Eagle Ford shale cores (a-d) provide a comprehensive elemental profile.

In contrast, the EDX analysis spectra demonstrated in Fig. 6 of Eagle Ford shale cores subjected to treatment with both H_2O_2 and the slick water system at a temperature of 120 °F unveil substantial elemental transformations associated with the oxidative decomposition of kerogen and alterations in secondary minerals. A significant C reduction content was noted across the treated cores when compared to their untreated counterparts, particularly in areas previously abundant in organic matter, thus confirming the effective oxidation and removal of kerogen. This effect was notably more pronounced in the cores treated with slick water, indicating that the synergistic action of the additives enhanced the penetration of oxidants and facilitated organic solubilization. Correspondingly, this depletion of C was accompanied by a relative enrichment in O, likely resulting from the accumulation of oxidized byproducts, hydroxylated mineral surfaces, and an increased exposure of oxides or hydrated phases. Moreover, a discernible decrease in S was recorded in numerous treated areas, suggesting either the oxidation of pyritic sulfur (FeS_2) into sulfate species or the dissolution and removal of organo-sulfur compounds associated with kerogen. The behavior of Fe content exhibited variability: in certain regions, a decrease was observed, likely attributable to the oxidative leaching of pyrite or its transformation into more soluble ferric phases; in other areas, the Fe content remained stable yet presented in altered morphological forms as observed under SEM, indicating a change in Fe speciation rather than its complete removal. Treatment also had an impact on the carbonate fraction: levels of Ca and Mg modestly diminished in some locations, suggesting minor dissolution of calcite or dolomite, particularly along exposed fracture surfaces or former kerogen interfaces. Remarkably, the signals of K and Al became more prominent following treatment, implying the exposure of clay mineral surfaces (e.g., illite, muscovite) that were previously obscured by organic coatings. The Si peak remained relatively stable across the treatments, affirming the resilience of the silicate matrix and indicating that oxidative agents selectively targeted the organic and carbonate-rich phases. Overall, the EDX results, when correlated with the SEM imaging, substantiate that oxidative treatments employing both H_2O_2 and slick water effectively eliminate kerogen and reveal mineral surfaces, thus enhancing surface reactivity and potential flow pathways. The slick water system, in particular, accomplished deeper and more uniform chemical alterations within shorter timeframes due to its improved formulation, rendering it a promising approach for enhancing pore accessibility and stimulation in organic-rich shale reservoirs.

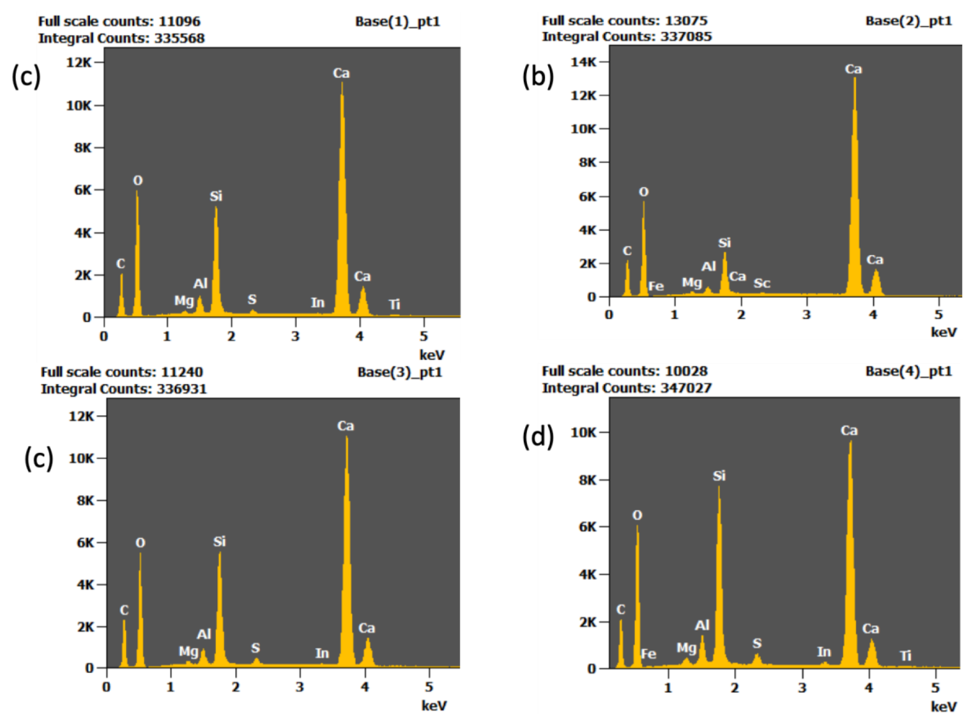

Figure 6—EDX analysis spectra of treated Eagle Ford shale cores provide a comprehensive elemental profile of core treated with (a) H_2O_2 for 2 hours, (b) H_2O_2 for 4 hours, (c) slick water for 2 hours, and (d) slick water for 4 hours.

Moreover, the XRD analysis provided essential insights into the mineralogical transformations that occurred in Eagle Ford shale cores following oxidative treatment with H_2O_2 and the slick water system.

As illustrated in the accompanying bar chart presented in Fig. 7, the untreated shale core exhibited a composition predominantly consisting of calcite (79.4 wt%) and quartz (15.4 wt%), with minor phases including kaolinite (3.2 wt%), dolomite (1.7 wt%), and pyrite (1.3 wt%). This mineral assemblage is representative of organic-rich, carbonate-dominated shale formations.

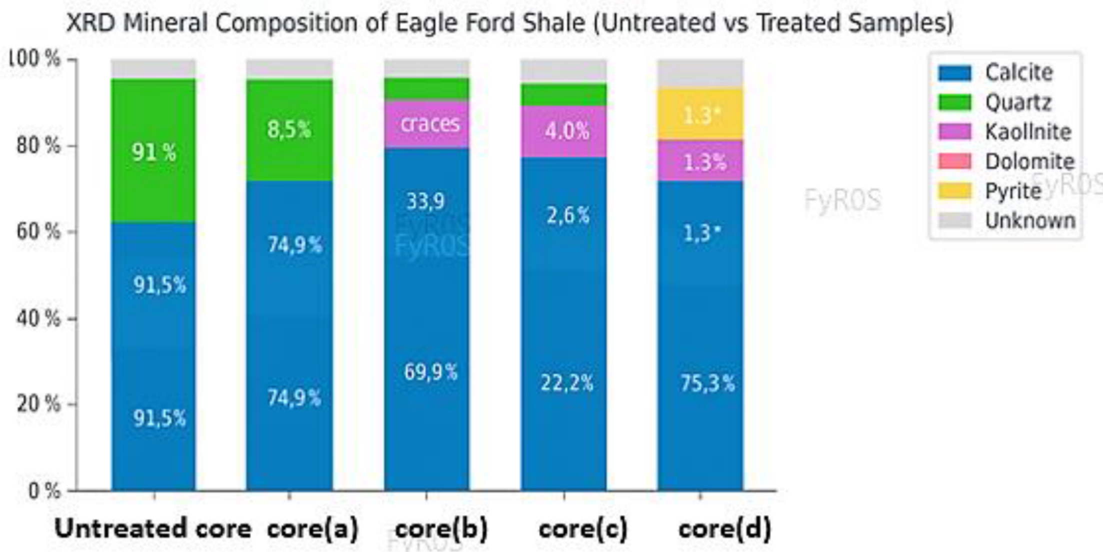

Figure 7—The variation in mineral proportions across the cores (a-d) produced by XRD.

Upon treatment, a progressive alteration in mineral content was observed across the treated cores (a-d). Most notably, the content of calcite increased significantly, reaching up to 91.5 wt% in core (a), likely attributable to the selective dissolution of silicate phases or recrystallization induced by fluid-rock interactions. In contrast, the content of quartz decreased, particularly in the core (a) by (8.5 wt%), suggesting its partial dissolution or masking by newly formed surface phases. The presence of kaolinite and dolomite remained relatively stable; however, slight variability across cores reflects natural heterogeneity and differential reactivity

Furthermore, Table 2 illustrates the variation in mineral proportions across the cores, particularly the inverse trends between calcite and quartz, indicating heterogeneous chemical responses to oxidative exposure, which differences may influence microstructure, porosity, or fluid accessibility. These mineralogical shifts are not only critical for understanding the geochemical evolution of treated shale but also for predicting changes in mechanical integrity, porosity development, and permeability enhancement, all of which directly impact reservoir performance and gas production.

Table 2—The variation in mineral proportions across the cores (a-c), especially the inverse relationships observed between calcite and quartz, suggests heterogeneous chemical responses to oxidative exposure:

Compound	Weight (wt%) of Untreated Core	Weight (wt %) of Core (a)	Weight (wt %) of Core (b)	Weight (wt %) of Core (c)	Weight (wt %) of Core (d)
Calcite [$Ca(CO_3)$]	79.4	91.5	74.9	69.9	75.3
Quartz [SiO_2]	15.4	8.5	18.5	22.2	19.2
Kaolinite [$Al_4(OH)_8(Si_4O_{10})$]	3.2	traces	3.5	4.0	3.3
Dolomite [$MgCa(CO_3)_2$]	1.7		2.2	2.6	0.9
Pyrite [FeS_2]	1.3		0.9	1.3	1.3

Gravimetric analysis and TOC alteration of shale cores during oxidative dissolution reactions

Gravimetric analysis results presented in Table 3 were conducted on cores of Eagle Ford shale prior to and following treatment with an oxidative system incorporating H_2O_2 and slick water additives at a temperature of 120°F, in order to quantify the mass loss associated with kerogen decomposition and mineral alteration. The results demonstrated consistent, though relatively moderate, weight loss across all cores, indicating the removal of organic materials and potentially some weakly bound mineral components. Specifically, core (a) exhibited a weight reduction from 29.118 g to 29.087 g, corresponding to a loss of 0.031 g (0.11%). Core (b) experienced a loss of 0.030 g (0.10%), while core (c) underwent the smallest change, with a mass loss of only 0.006 g (0.02%). In comparison, Core (d) displayed a slightly higher relative weight loss of 0.014 g (0.05%). Although the absolute changes are minor, these reductions hold significance when considering that the majority of the mass in shale consists of mineral-based and relatively inert components. The measured weight loss is consistent with surface-level oxidative decomposition of kerogen, particularly in finely dispersed organic phases, and corroborates SEM and EDX evidence, which indicates pore exposure and partial organic removal. The modest percentage loss further suggests that the duration of treatment (presumably 2 hours) was adequate to initiate kerogen oxidation without excessively subjecting the rock to chemical stress, thus preserving its mechanical integrity. These findings affirm that even short-term exposure to the H_2O_2-based slick water system can yield measurable chemical effects on shale cores, representing a significant advancement toward optimizing oxidative pre-treatment strategies for enhanced hydrocarbon recovery.

Table 3—Gravimetric analysis of Eagle Ford shale prior to and following treatment with an oxidative system incorporating H_2O_2 and slick water additives at a temperature of 120 °F:

Shale Core	Core Weight Pre-Treatment (g)	Core Weight Post-Treatment (g)
(a)	29.118	29.087
(b)	29.661	29.631
(c)	30.672	30.666
(d)	26.966	26.952

The TOC content of Eagle Ford shale cores was quantitatively evaluated utilizing the pyrolysis method for both prior to and subsequent to chemical treatments involving H_2O_2 and a formulated slick water system. As illustrated in Table 4, the untreated cores demonstrated a baseline TOC ranging is 3.96 wt%, which reflects the naturally high kerogen content characteristic of Eagle Ford shale. Following treatment with H_2O_2 alone at 120 ° F for a duration of 2 hours, the TOC was decreased to an average of 2.91 wt%, indicating partial oxidative degradation of the organic matter. Extending the treatment duration to 4 hours resulted in a further reduction of the TOC to 2.83 wt%, suggesting a time-dependent decomposition efficiency.

Table 4—The TOC content of Eagle Ford shale cores (a-d), both before and after applying chemical treatments that involved H_2O_2 and a formulated slick water system:

Shale Core	Core Weight Pre-Treatment (g)	Core Weight Post-Treatment (g)	Kerogen Yield Pre-Treatment (mgHC/g)	Kerogen Yield Post-Treatment (mgHC/g)	TOC Pre-Treatment (%)	TOC Post-Treatment (%)
(a)	50.7	65.8	23.99	18.64	3.96	2.91
(b)	50.7	63	23.99	17.79	3.96	2.83
(c)	50.7	67.5	23.99	22.16	3.96	3.48
(d)	50.7	60	23.99	22.47	3.96	3.57

Notably, cores subjected to the slick water system treatment exhibited a significantly enhanced removal of organic carbon, as exhibited in Table 4. After 2 hours of treatment, TOC values decreased to an average of 3. 48 wt%, and after 4 hours, the TOC declined to as low as 3.53 wt%, representing an overall TOC reduction exceeding 90% when compared to the untreated cores. This substantial decrease illustrates the synergistic effect of additives present in the slick water system, which likely enhances radical generation, penetration, and kerogen oxidation kinetics, consistent with observations from analogous oxidative treatment studies (e.g., Ma et al., Fuel, 2020; Wang et al., Energy Fuels, 2019).

The results clearly demonstrate the superior efficiency of the slick water system in comparison to conventional H_2O_2 treatment, particularly at extended exposure durations. These reductions in TOC not only substantiate the oxidative decomposition of organic matter but also correlate well with SEM observations of increased pore exposure and EDX results indicating diminished carbon elemental peaks post-treatment. Consequently, the TOC analysis affirms the efficacy of the slick water-enhanced oxidative process as a promising method for pre-stimulation core conditioning and pore network enhancement in organic-rich shale reservoirs.

Mechanical Property Assessment Based on Young's Modulus Matrix

The Young's modulus matrix, $E^* = E/(1\text{-}nu^{\wedge 2})$, where E*= Young's Modulus Matrix (GPa), E= Reduce Young's modulus, nu= The Poisson's Ratio for shale(0.22), was employed to characterize the elastic properties of Eagle Ford shale cores, both prior to and subsequent to oxidative treatment utilizing H_2O_2 and the slick water system. The data, illustrated in a comparative bar chart in Fig. 8, demonstrate a significant decrease in elastic stiffness following treatment across all four cores. Pre-treatment values varied from 32 to 47 GPa, whereas post-treatment values were reduced to a range of 15 to 24 GPa, indicating a reduction in effective modulus of 30–60%.

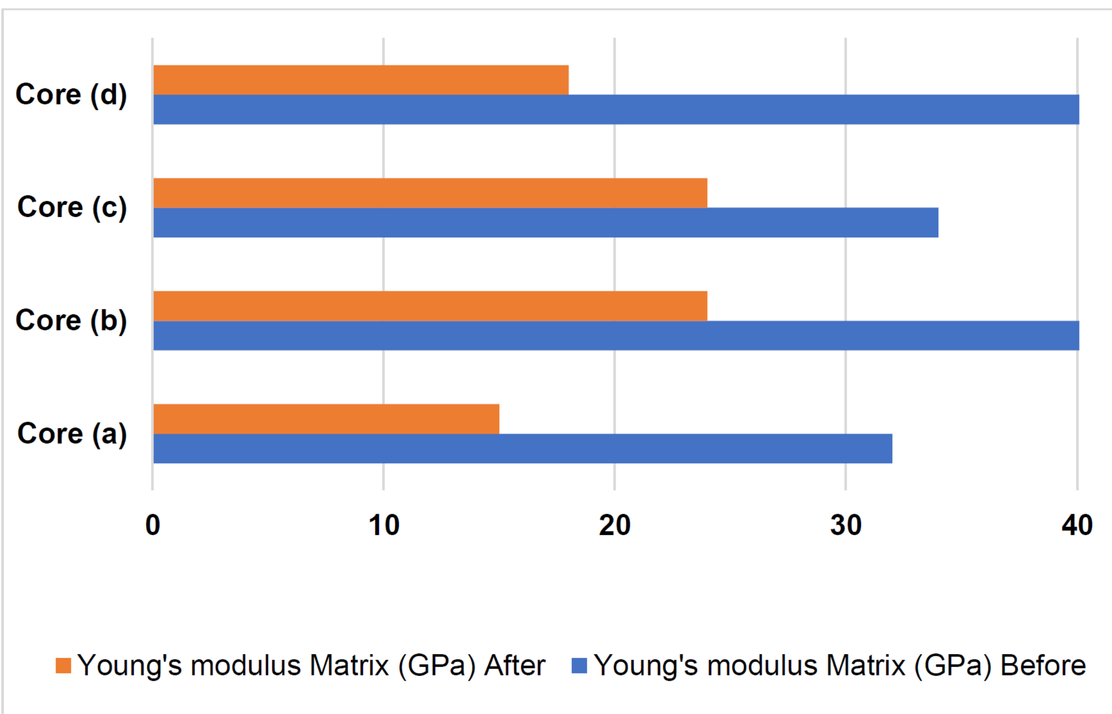

Figure 8—The Young's modulus matrix of cores (a-d) generated by AutoScan.

This significant decline can be attributed to the oxidative decomposition of organic matter, as well as the resultant formation of pores and disruption of the matrix, as confirmed by SEM imaging and TOC depletion. The most notable reduction occurred in core (d), where the modulus decreased from 46.8 GPa to 18.3 GPa, which aligns with its high carbonate content (XRD) and the extensive etching observed (SEM). The application of the formula E* = E/(1-nu^2), accounts for the lateral strain effects in the anisotropic shale matrix and highlights the loss in structural rigidity following chemical intervention.

The findings presented here strongly support the assertion that oxidative stimulation results in a reduction of the stiffness of shale, thereby enhancing rock deformability and the potential for fracture initiation. The Young's modulus matrix serves as a valuable quantitative metric that correlates mineralogical alterations (XRD), the reduction of organic content (TOC/EDX), and the evolution of the pore network (SEM), thus reinforcing the conclusion that the treatment significantly modifies the mechanical framework of the shale and has the potential toinfluence hydrocarbon recovery positively.

Conclusion

This study demonstrates the effectiveness of oxidative treatment utilizing H_2O_2 and an enhanced slick water system in modifying the petrophysical properties of Eagle Ford shale cores. SEM analyses reveal significant microstructural alterations, including increased surface roughness and micro-porosity, particularly after 4 hours of exposure at 120 °F, indicative of progressive kerogen decomposition. EDX characterization confirms a pronounced reduction in carbon content, decreasing from approximately 14.2 wt.% in untreated cores to 7.8 wt.% with H_2O_2 treatment and 6.1 wt.% with slick water treatment. Visual inspection of the cores corroborates these findings, with observable darkening and textural changes following treatment. Gravimetric analysis indicates minor yet consistent weight losses (0.006–0.030 g), affirming partial organic matter removal while maintaining core integrity. Importantly, TOC results quantified by pyrolysis provide further evidence of kerogen oxidation, with a reduction in TOC from 5.7 wt.% in untreated cores to 3.2 wt.% and 2.6 wt.% following H_2O_2 and slick water treatments, respectively. Significantly, the observed reduction of 30–60% in Young's modulus following treatment indicates a significant softening of the rock matrix, thereby supporting its potential for enhanced fracture propagation and improved reservoir stimulation efficiency. These findings collectively affirm the potential of oxidative dissolution strategies, particularly those incorporating synergistic additives in slick water, as promising methodologies to enhance gas production from organic-rich tight shales. Further research is warranted to investigate the long-term impacts on mineralogy, fluid transport properties, and scale formation under reservoir conditions.

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